From Prephysical Hypotheses to Physical Models

Conditional Uniform Additive Limit of Wilson–Dirac Relative Entropy on Bounded Scaled Masses

Kirtus Justus
Theoretical Physicist
Port Coquitlam, British Columbia, Canada

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Abstract

For interval restrictions of infinite-chain Wilson–Dirac occupied-band projectors with scaled mass $\mu = mL$, the fixed-positive-mass expansion of the relative entropy has logarithmic coefficient $\mu^2/6$ and an explicit finite part. This note shows that the additive remainder is uniform on bounded mass intervals $0 \le \mu \le M$, at rate $O_M(1/\log L)$, conditional on named finite-dimensional and trace-class estimates of the companion manuscripts, in particular the uniform boundary estimate (B9) of *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*. The additive bound does not establish a relative leading law along sequences of vanishing mass; a logical example shows why.

1 Additive and relative limits

This note records a consequence of the explicit estimates in the supplied Wilson manuscripts [2]. It is conditional on those estimates and their exact finite-dimensional and trace-class inputs. The argument below checks their uniform assembly; it does not independently certify every singular operator domain or boundary regularity proof on which the inputs depend.

The model is the restriction of an infinite-chain occupied-band projector, with Bloch Hamiltonian

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z.$$

Write $\mathcal{D}_L(\mu) = D(C_{L,\mu/L} \| C_{L,0})$, where C is the interval-compressed covariance, and set $N = 2L$. This is not the occupied projector of an open finite chain. The supplied fixed-positive-mass theorem states

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6} \log L + \mu^2 \left(\frac{13}{36} - \frac{\pi}{12} - \frac{\log \mu}{6} \right) - \mathcal{S}(\mu) + O_\mu(1/\log L).$$

Its error notation alone does not imply uniformity in μ. The finite inequalities behind it carry additional information.

Proposition 1 (Conditional uniform additive consequence)**.** *Fix $M > 0$. Assume the exact four-term decomposition below, Corollary 4 and equation (8) of* Uniform Modular Control and Nonlinear Convergence*, and the uniform boundary estimate*

(B9) of Cross-Boundary Trace-Class Estimates*, together with the exact two-bond formula stated below. Then, for a common sufficiently large* $L_0(M)$*,*

$$\sup_{0\le\mu\le M}\left|\mathcal{D}_L(\mu)-\frac{\mu^2}{6}\log L-\mu^2\left(\frac{13}{36}-\frac{\pi}{12}-\frac{\log\mu}{6}\right)+\mathcal{S}(\mu)\right|\le\frac{C_M}{\log L}\quad(L\ge L_0(M)).$$

At $\mu=0$*, the product* $\mu^2\log\mu$ *is defined by continuity, and* $\mathcal{D}_L(0)=\mathcal{S}(0)=0$*.*

Proof. Choose $L\ge\max(2,4M)$ and enlarge that common lower threshold when necessary. Then $N\ge\mu$, $r=\mu/(N+\mu)\le 1/2$, $L\ge 2\mu$, and $\log(1+\mu/L)\le 1/4$ throughout the mass interval. Thus the prerequisites used below have a common onset. Logarithmic estimates are applied first for $0<\mu\le M$ and extended to zero by continuity.

The exact decomposition is

$$\mathcal{D}_L=\mathcal{P}_L+\mathcal{Q}_N+\mathcal{F}_L-\Delta S_L.$$

We bound each contribution separately.

Modular contribution. Put $a_\mu=\sqrt{2/15}\,\mu^2/\pi$ and $b_\mu=\mu/(3\pi)$. Corollary 4 gives

$$|\operatorname{Tr}(S_NE_N)|\le\frac{14\mu/\pi+28a_\mu}{N}+\frac{112b_\mu}{N^2},$$

$$V_N\le\frac{392(H_N+1)}{\pi^2N^2}+49Ne^{-\pi N/6},\qquad|\operatorname{Tr}(D_N^{\mathrm{tr}}E_N)|\le 2\sqrt{\mu V_N/\pi}.$$

Here H_N is the harmonic number and $\mathcal{F}_L=\operatorname{Tr}[(S_N+D_N^{\mathrm{tr}})E_N]$. The coefficients are explicit polynomials in μ, while V_N is mass independent. Therefore

$$\sup_{0\le\mu\le M}|\mathcal{F}_L|=O_M(\sqrt{\log N}/N).$$

Nonlinear contribution. Let $c_\eta=11/18-\pi/12-(\log 2)/2$, and C_η be the finite mass-independent constant in equation (8) of the same paper. That inequality reads

$$\begin{aligned}|\mathcal{Q}_N-\mu^2c_\eta|\le{}&\frac{19}{3\sqrt{2}}NJ_1(r)K+\frac{2C_\eta}{35}N^2J_3(r)+\frac{2c_\eta\mu^3}{N}+\frac{\mu^2}{384K^3}\\&+\frac{\sqrt{2}}{K}\left(\frac{3a_\mu}{8}+\frac{3b_\mu}{2N}\right)+2C_\eta\sqrt{\left(\frac{a_\mu}{6N}+\frac{b_\mu}{2N^2}\right)M_{2,N}},\end{aligned}$$

where $M_{2,N}\le 32(H_N+1)/\pi^2$ and

$$J_1(r)\le\frac{r^2}{\pi}\left(1+\frac{1}{2}\log\frac{2}{r}\right),\qquad J_3(r)\le\frac{r^4}{\pi}\left(\frac{1}{2}+\frac{3}{8}\log\frac{2}{r}\right).$$

The elementary bound $\sup_{x\ge0}x^a\log_+(1/x)=1/(ae)$, $a>0$, and $\log(2/r)\le\log(3N)+\log_+(1/\mu)$ imply

$$\begin{aligned}NJ_1(r)&\le\frac{M^2[1+\frac{1}{2}\log(3N)]+1/(4e)}{\pi N},\\N^2J_3(r)&\le\frac{M^4[\frac{1}{2}+\frac{3}{8}\log(3N)]+3/(32e)}{\pi N^2}.\end{aligned}$$

Use the same $K = \lfloor \sqrt{N/\log N} \rfloor$ for every mass. For large N, $K^{-1} \le 2\sqrt{\log N/N}$. Every term is then bounded uniformly, and

$$\sup_{0\le\mu\le M} |\mathcal{Q}_N - \mu^2 c_\eta| = O_M(\sqrt{\log N/N}).$$

This uses the supplied finite all-degree estimate; it does not exchange uncontrolled mass and infinite-series limits.

Two-bond contribution. The exact formula in *Relative Entropy of Finite-Interval Wilson–Dirac Quasifree States*, pp. 3–4, is

$$\mathcal{P}_L = -\pi \left[\frac{N^2-1}{6} \left(u_m + v_m - \frac{2}{\pi} \right) + \frac{1}{2}(u_m - v_m) \right].$$

In the paper's elliptic parameter convention,

$$u_m + v_m = \frac{2}{\pi} \frac{\mathbf{E}(1-r^2) - r^2 \mathbf{K}(1-r^2)}{1-r^2}, \qquad u_m - v_m = \frac{2r}{\pi} \frac{\mathbf{K}(1-r^2) - \mathbf{E}(1-r^2)}{1-r^2}.$$

The complementary modulus in the convergent series of [1] is r. On $0 < r \le 1/2$, with $\ell = \log(4/r)$, those series give

$$\mathcal{P}_L = (N^2-1) r^2 \left(\frac{\ell}{6} - \frac{1}{4} \right) - r(\ell - 1) + O(N^2 r^4 \ell + r^3 \ell),$$

with absolute constants. This uniform remainder follows, for example, by bounding the remaining convergent power-series coefficients on $r^2 \le 1/4$; the logarithmic factors are retained explicitly. With $A = (N^2-1)r^2$ and $\ell_0 = \log(4N/\mu)$,

$$0 \le \mu^2 - A \le \frac{2\mu^3}{N} + \frac{\mu^4 + \mu^2}{N^2}, \qquad 0 \le \ell - \ell_0 \le \mu/N.$$

For each $a > 0$, $\mu^a \ell \le M^a \log(6N) + 1/(ae)$. The coefficient error, linear r term and elliptic remainders therefore give

$$\sup_{0\le\mu\le M} \left| \mathcal{P}_L - \frac{\mu^2}{6} \log L - \mu^2 \left(\frac{1}{6} \log \frac{8}{\mu} - \frac{1}{4} \right) \right| = O_M(\log L/L).$$

Boundary contribution and zero endpoint. Equation (B9) of *Cross-Boundary Trace-Class Estimates*, p. 7, states directly

$$\sup_{0\le\mu\le M} |\Delta S_L(\mu) - \mathcal{S}(\mu)| = O_M(1/\log L).$$

Its trace-path definition has $\|F'\|_\infty \le 2$, and (B7) gives the nuclear bound $\|\mathbb{D}_\mu\|_1 \le C\sqrt{\mu}$. Consequently $|\mathcal{S}(\mu)| \le C\sqrt{\mu}$ and $\mathcal{S}(0) = 0$ continuously. The finite covariances coincide at zero mass. This concerns a finite trace path, not subtraction of two divergent continuum entropies.

Adding the two scalar constants gives $13/36 - \pi/12 - (\log \mu)/6$. The boundary rate is slower than the other three error rates. Their sum proves the stated uniform additive bound. □

What does not follow. The proposition controls the absolute remainder for any bounded sequence μ_L, including sequences tending to zero. It does not establish

$$\mathcal{D}_L(\mu_L) \sim \frac{\mu_L^2}{6} \log L.$$

Even the additive error divided by that leading term is bounded only by

$$\frac{6C_M}{\mu_L^2 (\log L)^2},$$

which need not tend to zero. The mass-dependent finite part also requires relative control. For a purely logical example, the residual $r_L(\mu) = L\mu e^{-L\mu}/\log L$ vanishes at zero, tends rapidly to zero for each fixed positive mass, and is uniformly at most $1/(e \log L)$, yet at $\mu = 1/L$ it exceeds the scale $L^{-2} \log L$ by an unbounded factor. This example concerns the validity of an inference; it is not asserted to be the actual Wilson residual.

Status of the dependencies. The inspected modular Riccati estimates, Jacobi edge comparison, all-degree path argument and positive-weight manipulations are coherent with the displayed bounds. The exact concentration connection and boundary theorem remain substantive analytic dependencies. The present work has not independently reproduced every domain and regularity argument. Accordingly, Proposition 1 is stated with its inputs explicit. It corrects an overbroad reading of the phrase "no uniform shrinking-mass theorem follows" without claiming a relative asymptotic or an optimal rate.

Research and computational disclosure

This is an AI-assisted mathematical research manuscript prepared within the same research program as the cited companion manuscripts. The argument is presented for examination. It checks the uniform assembly of the cited estimates; it does not independently certify those estimates, and it does not constitute independent peer review. Mathematical, computational and editorial assistance: ChatGPT (OpenAI). Editorial preparation of this edition: Claude (Anthropic).

References

[1] NIST Digital Library of Mathematical Functions, section 19.12, equations 19.12.1–19.12.3, convergent complementary-modulus series. https://dlmf.nist.gov/19.12 (accessed 10 November 2026).

[2] K. Justus, *Finite Part of Wilson–Dirac Relative Entropy*; *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*; *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*; and *Relative Entropy of Finite-Interval Wilson–Dirac Quasifree States*, supplied manuscripts (2026). Exact locations are identified in the text. These sources share research provenance.